Hyperspectral Image Super-Resolution via Intrafusion Network

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***Abstract*— This article presents an intrafusion network (IFN) for hyperspectral image (HSI) super-resolution (SR). Given that the HSI is a 3-D data cube with both the spatial information and the spectral information, the key challenge to construct HSI SR is how to efficiently exploit the spectral information among consecutive low-resolution (LR) bands, besides the spatial information. The proposed IFN consists of three modules, including the spectral difference module, the parallel convolution module, and the intrafusion module, which directly utilizes both the spatial information and the spectral information for reconstructing the high-resolution HSI. Different from most of the existed methods that tackle the spatial and spectral information separately, the proposed spatial–spectral utilization is achieved in one integrated network, which opens up a new way for HSI SR. Meanwhile, applications of this three modules strategy (first spectral difference, then parallel convolution, and finally, intrafusion) on both the conventional convolutional neural network and the residual network with deeper depth have shown the generalization capacity of this proposal. Experimental results and data analysis demonstrate the effectiveness of the proposed method using three hyperspectral data sets.**

***Index Terms*— Hyperspectral image (HIS), intrafusion, spectral difference, super-resolution (SR).**

I. Introduction

HYPERSPECTRAL imagery sensors usually collect hundreds of bands from the visible to the infrared wavelength (400–2500 nm). Due to the high spectral resolution, hyperspectral images (HSIs) with the material discriminative ability are suitable for many applications, including target detection [1], land cover mapping [2], and change detection [3]. However, limited by the optical imagery system, there is a tradeoff between the spectral resolution and the spatial resolution of the HSIs. The spatial resolution of the HSIs is sacrificed for their spectral resolution. HSI super-resolution (SR) aims at enhancing the spatial resolution of the input HSIs as postprocessing (without modifying the sensors), which is important for accurate classification and has become a hot issue in computer vision.

There are multiple studies on SR algorithms for HSIs. According to the number of input images, the existed work mainly proceeds into two categories. One is with fusion operation, with more than one input image. The other is just a single-to-single SR process.

Concerning the fusion approach, it is based on the assumption that some auxiliary images of the same scene are known. Li *et al.* [4] suggested that a hidden Markov tree can be applied to set up a prior model from the wavelet domain for a few low-resolution (LR) images, and the maximum *a posteriori* (MAP) method is adopted to calculate the high-resolution (HR) image and to update the prior model. Dian and Li [5] and Dian *et al.* [6] proposed to estimate a low-dimensional spectral subspace from the input LR HSI, and the coefficients are estimated with low tensor multirank prior from another HR multispectral image (MSI) [low-tensor train rank (LTTR)]. Li *et al.* [7] discussed that the input LR HSI can first be super-resolved by utilizing a network that is designed for normal images. Meanwhile, the obtained pre-HR HSI can be further fused with both the input LR HSI and the HR MSI. Dong *et al.* [8] proposed a nonnegative structured sparse representation (NSSR) approach to jointly estimate the dictionary and sparse code of the HR HSI based on the input LR HSI and HR RGB image. Fang *et al.* [9] learned a spectral dictionary from the LR HSI and transformed the dictionary for the HR MSI. Moreover, MSI is clustered into superpixels, and similarities for the pixels in the cluster are utilized to improve the sparse decomposition and to achieve a more accurate fractional abundance coefficient. In this way, the HR HSI can be reconstructed by the spectral dictionary and the fractional abundance coefficient matrix. With another image as the input, more information is input into the super-resolving process, making this type of methods often achieve a better spatial enhancement. However, in reality, auxiliary coregistered images with the same scene for the input LR HSI are usually hard or impossible to be accessed, which limits the practicability of this type of method.

When comes to the single-to-single approach, it is based on just one input LR HSI. This category can be further divided into two subgroups. One is with subpixel mapping based on spectral unmixing results. Since HSIs are always with coarse spatial resolution, the pixels in the HSIs are usually

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mixed by several end-members [10]. Spectral unmixing aims at estimating the fractional abundance of pure ground objects within a mixed pixel, and they can be used for subpixels classification. Zhao *et al.* [11] utilized conditional random fields to make full use of the input fraction image and alleviated the effect of errors brought during the spectral unmixing process. Arun *et al.* [12] explored the convolutional neural network for subpixel mapping the HSIs. This network essentially treats the subpixel mapping in the single image SR subdirection. Both the subpixel mapping process and the spectral unmixing process are integrated into one network, in which the finer scale classified maps are used for the training labels. Li *et al.* [13] proposed a framework that utilized both the spectral mixture analysis and the spatial–spectral sparsity. This kind of method tackles the SR problem from the end-member extraction and fractional abundance estimation. For this type of method, the noise generated by the unmixing operation is evitable during the mapping operation, which makes a negative influence on the SR process [14]. In addition, subpixel mapping methods are usually applied to certain applications, such as classification and target detection, for overcoming the limitation in spatial resolution [15].

The other subgroup is based on super-resolving the HSIs band by band, in which the HSIs are mainly processed from their spatial aspect. This type of method recovers the subpixel information mainly based on the LR-HR mapping that is learned from millions of training data and labels. Xie *et al.* [16] proposed to super-resolve some key bands via a deep neural network first and then to super-resolve the rest bands according to their feature matrix. Meanwhile, to make a further utilization of the spectral information, [7], [17], [18] all utilized a coupled nonnegative matrix factorization (CNMF) [19] to make a further fusion, which obtained finer HR outputs. Before utilizing the CNMF to make a further fusion, [17] transferred the normal deep learning network (TrsfrLea) designed for the color images to each band of the HSIs. Instead of super-resolving the HSI band by band, [18] utilized an intrainterpolation operation to get a coarser HR HSI, and the CNMF is utilized to obtain a finer one, which makes a deep spatial and the spectral exploration (DSSE) of the information the LR HSI conveys. Meanwhile, this operation that first super-resolves the bands just by their spatial information and then incorporates a manual way to make a further utilization of the spectral information is inflexible and unstable. Meanwhile, the performance of the CNMF highly depends on the iteration times, number of the end-members, and so on.

In this article, we present a new approach that opens up a new way to enhance spatial resolution via spatial–spectral exploitation. Our main contributions are threefold.

First, the proposed intrafusion network (IFN) takes the current band and its neighboring LR band as the input, and the resultant HR estimate is automatically aggregated according to the IFN. Due to the design of the spectral difference module, the IFN is able to robustly extract not only the spatial information but also the useful information among consecutive bands. Both the extracted spatial and spectral information can be applied to reconstruct the HR images, an operation that ensures acceptable performance.

Second, besides utilizing both the spatial and the spectral information for the SR process, the IFN also conducts the extraction in one integrated network. By importing a parallel convolution module, feature maps of the current band, its neighboring band, and their spectral difference are learned simultaneously. Different from most of the existed methods that utilize the spatial and spectral information separately, the IFN only consumes little manual manipulation to achieve information fusion in a subtle way.

Third, the spectral difference module developed in the IFN is a general and effective module for improving the SR performance of the HSIs and can directly be applied to a wide range of other networks. This module benefits from automatic spectral information utilization. Performance on this module embedded in different network architectures has been systematically investigated and thoroughly analyzed, and the effectiveness of this module has been proved.

The rest of this article is organized as follows. Section II describes the proposed IFN. Experimental results and data analyses are presented in Section III. Section IV presents the conclusion.

II. IFN

In this section, some important observations are first introduced, and a detailed description of the architecture of the proposed IFN is described. The architecture of the IFN can be further divided into three parts: 1) spectral difference module; 2) parallel convolution module; and 3) automatic aggregation module.

A. Important Observations

The design of the IFN is based on two specific insights. Li *et al.* [20] showed that spectral difference can be used for reconstructing the HR HSI in an efficient and effective way via the spectral difference convolutional neural network (SDCNN). However, it is noted that the SDCNN just learns the mapping between the LR spectral difference and the HR spectral difference. When super-resolving one band, the learned spectral difference is combined with the former reconstructed HR bands. The super-resolving process is separated from the mapping learning process. This inspired the design of integrating the spectral difference module with the super-resolving process in one architecture.

The second insight comes from the SR convolutional neural network (SRCNN) [21]. SRCNN is the pioneer of applying deep learning to normal RGB image SR. SRCNN is designed with a lightweight structure and has achieved appealing performance. Inspired by the SRCNN, the IFN is also first designed with three convolutional layers to validate the effectiveness of the IFN with three modules.

B. Network Architecture

In order to facilitate discussion, some notations of the frequently used terms are clarified. The input LR HSI and

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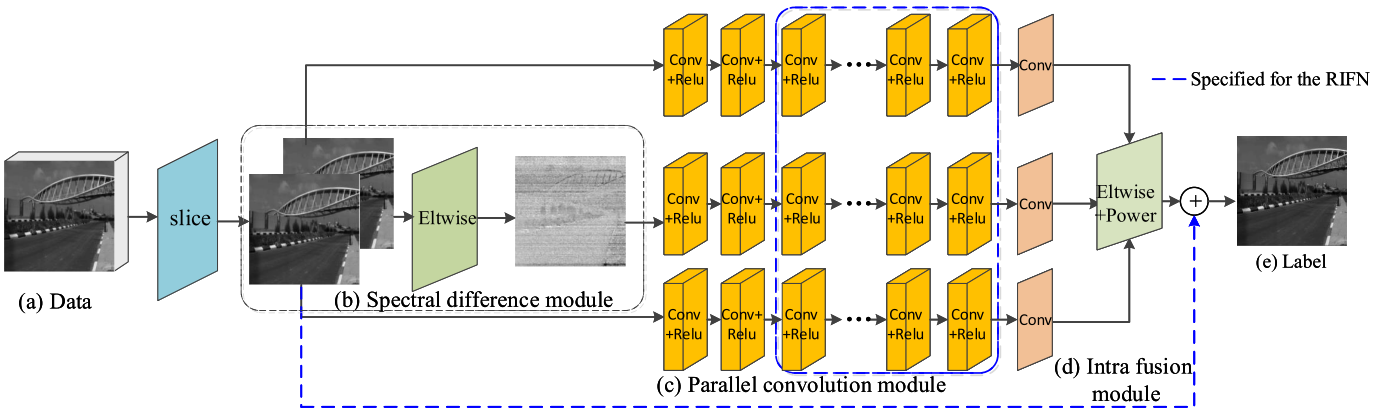


Fig. 1. Network architecture of the proposed IFN and the RIFN. (a) Data. (b) Spectral difference module. (c) Parallel convolution module. (d) Intrafusion module. (e) Label.

the desired HR HSI are denoted as $\mathbf{L} \in R^{w\times h\times b}$ and $\mathbf{H} \in R^{\mathrm{sw}\times \mathrm{sh}\times b}$, where w and h represent the width and the height of the LR HSI. n and s denote the number of bands in the HSI and the scaling factor, respectively.

Inspired by the spectral difference module and the architecture of the SRCNN, the proposed IFN is designed, as shown in Fig. 1. In the spectral difference module, it is noted that two adjacent LR bands act as the input. A slice layer is first designed to split these two bands, and then, an element-wise (eltwise) layer is followed to obtain their spectral difference. In the parallel convolution module, three branches are designed for the current band, its neighboring band, and their spectral difference, respectively. For each branch, three convolutional layers are utilized to extract the feature maps, which is the same manipulation as that in the SRCNN. Moreover, in the automatic aggregation module, feature maps of these three branches are added in the final part of the IFN. Finally, a loss layer is followed to generate the reconstructed HR image. In this way, during the training process, the optimization problem between the target HR band and the predicted output can be denoted as

$$\min_{\Theta} \sum_{j} \|F(\boldsymbol{L}_j, \boldsymbol{L}_{j+1}; \Theta) - \boldsymbol{H}_j\|_2^2 \tag{1}$$

where $F(\boldsymbol{L}_j, \boldsymbol{L}_{j+1}; \Theta)$ represents the output of the IFN. $\boldsymbol{H}_j$ is the jth band of the desired HSI with high spatial resolution, and j ranges from 1 to $b-1$. $\boldsymbol{L}_j$ and $\boldsymbol{L}_{j+1}$ are the associated LR bands. Θ is the set of parameters in the network.

1) Spectral Difference Module: Each band is processing separately. For the ith band, spectral difference image between the ith and the $(i+1)$th band is the pixelwise subtraction result of these two bands. Formally, given an image, $\boldsymbol{L}_i$, and its neighboring band $\boldsymbol{L}_{i+1}$, the spectral difference module computes the pixel difference $\boldsymbol{D}_i(x, y)$ for every pixel location (x, y) in the band $\boldsymbol{L}_i$. That is

$$\boldsymbol{D}_i(x, y) = \boldsymbol{L}_{i+1}(x, y) - \boldsymbol{L}_i(x, y) \tag{2}$$

where i ranges from 1 to $b-1$. Note that this module maps the spectral difference from the present bands to the latter band. The current band, its neighboring band, and their spectral difference band are used as the input to the next module.

2) Parallel Convolution Module: This module works on three bands generated from the spectral difference module. Three branches are designed for each band, respectively. The pioneer neural network for normal image SR, i.e., SRCNN, is customized to validate the effectiveness of this IFN. There are three layers for each branch. The first layer is a convolutional layer with 64 kernels in the size of 9×9, the second layer is another convolutional layer with 32 kernels in the size of 1×1, and the last layer is also a convolutional layer with one kernel in the size of 5×5. The rectified linear unit (ReLU) [22] is chosen as the activation function of the first and the second layers. All the convolutions are conducted with stride one.

Afterward, a deeper residual network with more trainable parameters acts as the parallel convolution module and is developed to further prove the significance and innovation of our idea. The very deep residual network (VDSR) [23] has been applied to the SR for the color images. In this way, the three branches in the intrafusion can also be designed by this residual network. To make a distinction between the IFNs whose parallel convolution modules are different, the IFN that is inspired by the residual network is named as the residual IFN (RIFN). There are 20 convolutional layers in the VDSR. Each convolutional layer has 64 kernels in the size of 3×3 with stride one and a ReLU as the activation function, where a filter operates on a 3×3 spatial region across 64 channels (feature maps). The RIFN is a further application of the VDSR in the HSI super-resolving process, and all the parameters of the three branches in the parallel convolution module are the same as those of the VDSR. The workflow of the RIFN is also exhibited in Fig. 1. The convolution layers in the grid and the arrow that is denoted by the blue dotted line are specified for the RIFN. As shown in Fig. 1, the RIFN differs from the IFN in the target of the learning process and the convolution layers. Residual images are learned via the network RIFN, an operation that greatly improves the convergence speed. Meanwhile, more convolution layers are utilized in the RIFN. The RIFN learns residual images via the network, an operation that greatly improves the convergence speed. Note that the

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design of the parallel convolution module is not limited to these two choices, and most other networks designed for normal image SR, such as efficient sub-pixel convolutional network (ESPCN) [24] and Information Distillation Network (IDN) [25], can work as the parallel convolution module as well. The output of these three branches serves as an estimate for the final HR band.

3) Automatic Aggregation Module: The principle of this module is to combine the outputs of the parallel convolution module in a pixelwise manner. Specifically, the sum layer is designed to collect the feature maps of the current band, its neighboring bands, and the spectral difference, which makes an intrafusion in the feature map perspective. The output feature maps of the parallel convolution module are summed up to form the final estimated HR band, which can be denoted as

$$\frac{F_{p1}(\boldsymbol{L}_j;\Theta_{p1})+F_{p2}(\boldsymbol{D}_j;\Theta_{p2})+F_{p3}(\boldsymbol{L}_{j+1};\Theta_{p3})}{2} \tag{3}$$

where F_{p1}, F_{p2} and F_{p3} are the output of the three branches in the parallel convolution module, respectively. Θ_{p1}, Θ_{p2}, and Θ_{p3} represent the parameters of each branch. In this way, the optimization problem modeled by (1) can be described in a more detailed format

$$\min_{\Theta}\sum_{j}\left\|\frac{F_{p1}(\boldsymbol{L}_j;\Theta_{p1})+F_{p2}(\boldsymbol{D}_j;\Theta_{p2})+F_{p3}(\boldsymbol{L}_{j+1};\Theta_{p3})}{2}-\boldsymbol{H}_j\right\|_2^2. \tag{4}$$

It is noted that both the feature map extracting and the intrafusion operations are optimized simultaneously in an automatic and end-to-end mode.

III. Experimental Setup and Data Analysis

An experimental discussion is presented in this section. Data sets utilized for validating the performance are first introduced, and then, the implementation details are provided. Finally, a thorough analysis of the experimental results is provided.

A. Data Sets

In the experiment, the proposed method is evaluated on three synthetic data sets generated from real hyperspectral data, which are listed as follows.

1) *ICVL Data Set:* The first data set was created by the International Computational Vision Laboratory (ICVL). The ICVL data set contains 201 images until now and will continue to grow progressively. All the HSIs in the ICVL data set were acquired using a Specim PS Kappa DX4 hyperspectral camera and a rotary stage for spatial scanning. These HSIs were collected with the spatial resolution of 1392 × 1300 over 519 spectral bands in the.raw format. The volume of each HSI is almost 2 GB, which is a great burden for the training process. For researchers' convenience, the ICVL laboratory also provided a.mat version of these HSIs via the online link,[1] and these.mat files were downsampled to 31 spectral channels from 400 to 700 nm at 10-nm increments. Moreover, among these 201 HSIs, many are highly overlapped and were captured under the same scene with just 1- or 2-min interval, for example, the HSIs named "objects_0924-1622" and "objects_0924-1623." The suffices indicate that both HSIs were captured on 24th September with just 1-min interval. Their RGB descriptions are exhibited in Fig. 2.

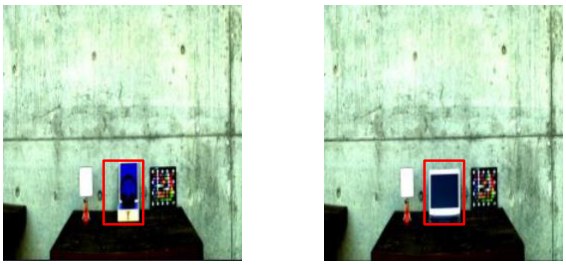


Fig. 2. Color image of two HSIs from the ICVL database, which were captured just with 1-min interval.

It is noted that most of the scenes in the images are the same. We have incorporated the correlation coefficient (CC) measurement to determine the correlation between these two HSIs. Among all the bands, the largest CC is 0.9122, and the smallest is 0.8718. The mean value of the CCs for all these bands is 0.8957. Furthermore, we have eliminated the red rectangle region in these two HSIs to compute the CCs. The maximum, the minimum, and the mean value of the CCs are 0.9761, 0.9343, and 0.9416, respectively. Both the subjective visual exhibition and the objective quantitative measurement have demonstrated the high similarity between these two HSIs. In this way, to balance the scene diversity and the capacity of the platform, we finally adopt 48 HSIs in the.mat format for validating the performance. These 48 HSIs describe different scenes and are with both rich and comparatively simple spatial texture. The numbers of the HSIs in the training set, validation set, and the test set are 32, 8, and 8, respectively.

2) *Pavia Center:* The second HSI was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) during a flight campaign over Pavia, Northern Italy. The Pavia Center is with 1096 × 1096 pixels in the spatial aspect, but some of the samples contain no information and have been discarded before analysis, leaving 1096 × 715 pixels. It was initially composed of 115 bands that have been reduced to 102 bands after removing the water vapor absorption and noise bands.

3) *Washington DC Mall:* This data set was taken over the National Mall in Washington, DC, USA, in 1995. It was acquired by the Hyperspectral Digital Image Collection Experiment (HYDICE), which measures pixel response in 210 bands in the 0.4–2.4-μm region of the visible and infrared spectrum. Bands in the 0.9- and 1.4-μm region where the atmosphere is opaque have been omitted

[1] http://icvl.cs.bgu.ac.il/hyperspectral/

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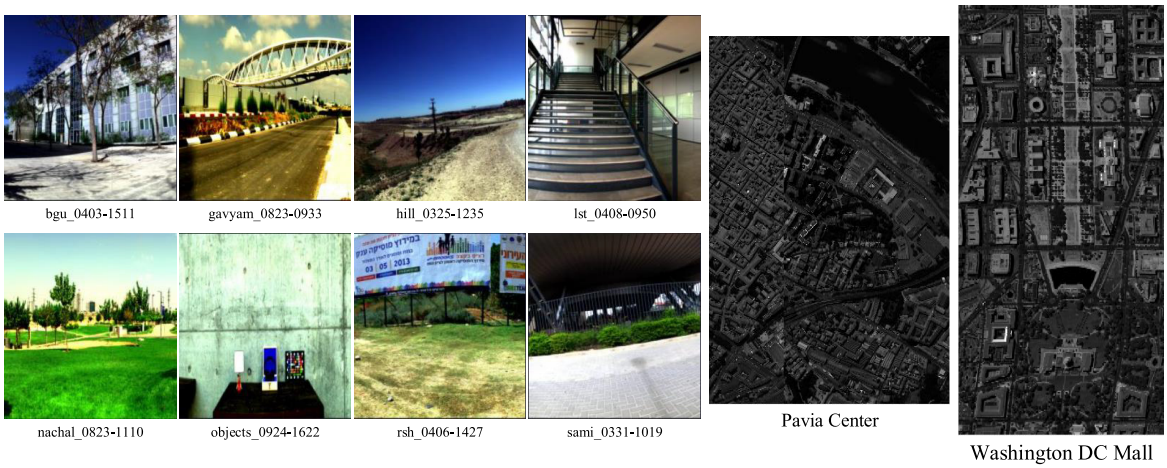


Fig. 3. Color image of the test HSIs from the ICVL database and gray images of the other two airborne remote sensing HSIs.

from the data set, leaving 191 bands. This data set contains 1208 scan lines with 307 pixels in each scan line. In the experiment, the image that is widely utilized in target detection has been exploited, and its size is $640 \times 307 \times 191$.

Color images of the eight test HSIs from the ICVL data set are exhibited in combination with the gray images of the other two HSIs in Fig. 3.

B. Experimental Setup

For each ground-truth image, the LR input is downsampled via the bicubic interpolation method by a ratio of 4. In the training process, the ground-truth images are randomly cropped into subimages whose size is $f_{\text{is}} \times f_{\text{is}}$. The LR samples are synthesized by the first downsampling by the scaling factor. Then, the downsampled samples are upscaled by the same factor via bicubic interpolation. To avoid border effects during training, all the convolutional layers have no paddings, and a smaller output whose size is $(f_{\text{os}} \times f_{\text{os}})$ is produced. The MSE loss function is evaluated by the difference between the central pixels of the ground-truth images and the output of the network. It is noted that for the Pavia Center and the Washington DC Mall, the data utilized for the training are far less than those from the ICVL data set. In this way, to enrich the amount of the training data for these two data sets, f_{is} for them is 33 and is smaller than 63 for the ICVL data set. The corresponding f_{os} changes with f_{is}. Specifically, the Pavia Center is composed of two regions, which are 223 columns and 492 columns, respectively. To avoid the disturbances caused by the discontinuity between these two regions, the left region with 223 columns is utilized for the test set. The first 400 columns of the right region (from the 224th to the 623rd ones) and the rest 92 columns of the right region (from the 623rd to the 715th ones) act as the training set and the validation set, respectively. In the case of the Washington DC Mall HSI, it is demonstrated in Fig. 3 that the scene in the horizontal direction is more homogeneous than that in the vertical direction. Scenes at different rows are comparatively more heterogeneous. Therefore, to balance the scene diversity in the training, validation, and testing set, the Washington DC Mall is also separated from the vertical direction. The first 203 columns, the 204th to the 255th columns, and the rest 52 columns act as the training set, the validation set, and the testing set, respectively.

To validate the superiority of the proposed method, a simple but effective IFN with three layers is designed. Its architecture is shown in Fig. 1. Comparison is made between the typical three-layer network SRCNN. To make a fair comparison, it is noted that both the number and the size of kernels at each layer for the IFN and the SRCNN are the same. Meanwhile, to make a comprehensive comparison, this intrafusion strategy is also applied to the residual network to get a deeper network RIFN for the Pavia Center and the Washington DC Mall. The architecture of the RIFN is also demonstrated in Fig. 1. The same spectral difference and intrafusion modules are applied to the IFN and RIFN, and the parallel convolution modules are different. The RIFN is inspired by the residual network VDSR with more convolution layers. In this way, the comparison is also made between the RIFN and the VDSR. Any branch of the parallel convolution module in the RIFN is the same as the VDSR.

The following six quantitative measurements, root-mean-square error (RMSE), CC, spectral angle distance (SAD), the Erreur Relative Globale Adimensionnelle De Synthese (ERGAS) [26], peak signal-to-noise ratio (PSNR), and structure similarity index measurement (SSIM), are employed to evaluate the performance of reconstructed HR HSIs. RMSE, CC, PSNR, and SSIM are universal measurement matrices for image quality assessment, and their detailed descriptions are omitted in this article. SAD evaluates the spectral preservation for all the pixels. The SAD at the kth pixel is determined by

$$\text{SAD}(s_k, \hat{s}_k) = \arccos\left(\frac{\langle s_k, \hat{s}_k \rangle}{\|s_k\| \|\hat{s}_k\|}\right) \tag{5}$$

in which $\langle s_k, \hat{s}_k \rangle = {s_k}^T \hat{s}_k$ is an inner product between the spectral curves s_k and $\hat{s}_k$, and $\|.\|$ is the l_2 norm. The optimal value of SAD is 0. The values of SAD reported in our experiments are obtained by averaging the values obtained for all the image pixels. ERGAS is a global statistical measure of the SR quality whose best value is also 0. The ERGAS of H

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and $\hat{H}$ is calculated by

$$\mathrm{ERGAS}(H, \hat{H}) = 100s\sqrt{\frac{1}{b}\sum_{k=1}^{b}\left(\frac{\mathrm{RMS}E_k}{\mu_k}\right)^2} \tag{6}$$

in which $\mathrm{RMS}E_k = (\|\hat{H}_k - H_k\|_F/\sqrt{n})$, and μ_k is the sample mean of the kth band of ground truth H.

Among these six indices, the larger the CC, PSNR, and SSIM are, the better the quality is. The rest RMSE, SAD, and ERGAS are exactly the opposite. All experiments were run on a desktop with Intel Core i5 2.8-GHz CPU and 16.0-GB RAM using MATLAB (R2015b). The training is operated on a platform with Intel Core i7 3.5-GHz CPU, 16-GB memory, and NVIDIA GTX 1080Ti. For the ICVL data set, it takes about 258.3 h to train the IFN with 5 000 000 times.

It should be noted that as the proposed intrafusion strategy can be applied to most neural networks and to achieve a better spatial information enhancement, the SRCNN and VDSR are two typical different network architectures that are modified to prove this inference. It is foreseeable that this strategy can be further applied to some other networks and also obtain better spatial information. Hence, when making a performance comparison, the difference between the normal network and the network with intrafusion is focused. Meanwhile, the comparison is made between the proposed method and some other state-of-the-art HSI SR methods, including the NSSR [8], LTTR [6], TrsfrLea [17], and DSSE [18]. Specifically, both the NSSR and the LTTR belong to the fusion approaches, and the rest two belong to the single-to-single ones. Both the NSSR and the LTTR methods require an auxiliary RGB image as the input, besides the LR HSI. To ensure different methods share the same input, the HR RGB image required is interpolated to the desired size by the LR RGB images [30]. For the HSIs from the ICVL data set, their spectral response transform matrices are provided to compute the RGB images. For both the Pavia Center and the Washington DC Mall, their RGB images are inaccessible. In this way, the 30th, 60th, and 90th bands are randomly selected and act as the three channels for the RGB images. Meanwhile, when super-resolving the last band in the HSI, it is duplicated and acts as the adjacent band for itself.

C. Data Analysis

In this section, an analysis of the number of convolution layers in the IFN is first made. Then, the performance of the proposed method and comparison with some other state-of-the-art methods are investigated.

In [21], it is demonstrated the SRCNN with three layers achieves better performance than the SRCNNs with both the four and five layers. In the case of the IFN, the architecture of the parallel convolution module is inspired and inherited from the SRCNN. To make a further validation of the parameter setting for the IFN, we have investigated the performance of the IFNs with both four and five layers and make a comparison with the IFN with three layers. Performance has been evaluated on both the "sami_0331-1019" HSI from the ICVL data set and the Washington DC Mall. Convergence curves in Fig. 4(a) depict that IFNs with deeper

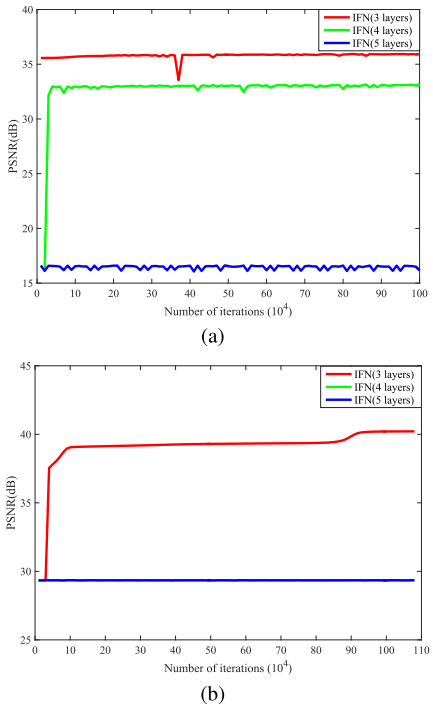


Fig. 4. Comparisons between IFNs with three layers, four layers, and five layers for (a) "sami_0331-1019" HSI and (b) Washington DC Mall.

convolution layers. It is noticed the IFNs with four and five layers do not improve the performance for the "sami_0331-1019" HSI from the ICVL data set. It may be caused by the difficulty of training. The proposed IFN contains no pooling layer or full-connected layer; thus, it is sensitive to the initialization parameters and learning rate. Once going deeper (e.g., four or five layers), it is hard to set appropriate learning rates that guarantee convergence [21]. Curves shown in Fig. 4(b) have clearly proved this divergence. These curves depict the PSNRs for the Washington DC Mall with different layers. Curves for the IFNs with four and five layers are approximately overlapped and have been exhibited in divergent states with unacceptable performance. The training samples for this data set are far less than those of the ICVL data set, making the IFNs with the deeper depth for this data set inclined to the divergent state. In this way, the parallel convolution module in the IFN is also designed with three layers.

Table I has listed the quantitative measurement metrics of the proposed IFN, with comparison to both the SRCNN and some other state-of-the-art HSI SR methods. The RMSE, CC, SAD, PSNR, and SSIM values are the averages of all band images and pixels, respectively. The best values are labeled in bold. It is noted that the proposed IFN always outperforms the SRCNN in both the spatial measurement PSNR and the spectral measurement SAD. Moreover, the proposed IFN always achieves the best SAD for all the test HSIs. The spectral difference module not only imports the spectral information but also constraints the spectral correlation in the super-resolving process and achieves better spectral measurement.

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TABLE I
PERFORMANCES OF DIFFERENT SR METHODS ON THE EIGHT TEST HSIS FROM THE ICVL DATA SET AT THE SCALING FACTOR OF 4

image	index	bicubic	NSSR [8]	LTTR [6]	TrsfrLea [17]	DSSE [19]	SRCNN [22]	IFN
bgu_0403-1511	RMSE	0.0149	0.0270	0.0170	0.0130	0.0127	0.0123	**0.0121**
	CC	0.9935	0.9852	0.9913	0.9950	0.9948	0.9956	**0.9957**
	SAD	0.9164	3.3188	1.4000	0.9987	1.0575	0.8693	**0.8476**
	ERGAS	1.4942	3.2738	1.6796	1.2918	1.2967	1.2279	**1.2088**
	PSNR	36.5605	31.3743	35.4074	37.6979	37.9579	38.2259	**38.3769**
	SSIM	0.9267	0.8475	0.9181	0.9424	**0.9522**	0.9428	0.9451
gavyam_0823-0933	RMSE	0.0145	0.0229	0.0154	0.0135	**0.0103**	0.0127	0.0126
	CC	0.9873	0.9785	0.9861	0.9895	**0.9938**	0.9903	0.9905
	SAD	0.6584	3.1301	0.9282	0.9069	0.9176	0.6471	**0.6356**
	ERGAS	1.8821	3.3460	1.9674	1.7197	**1.3408**	1.6342	1.6206
	PSNR	36.7444	32.8068	36.2289	37.4142	**39.7563**	37.9455	38.0166
	SSIM	0.9439	0.8605	0.9410	0.9509	**0.9671**	0.9513	0.9525
hill_0325-1235	RMSE	0.0127	0.0208	0.0133	0.0129	0.0125	0.0117	**0.0117**
	CC	0.9900	0.9783	0.9895	0.9904	0.9910	0.9915	**0.9916**
	SAD	0.7646	3.6560	0.9542	1.0310	1.0192	0.7554	**0.7524**
	ERGAS	1.8275	2.9150	1.8777	1.7884	1.7311	1.6809	**1.6799**
	PSNR	37.9073	33.6358	37.5247	37.8175	38.0896	38.6135	**38.6259**
	SSIM	0.9274	0.8564	0.9251	0.9383	**0.9406**	0.9387	0.9386
lst_0408-0950	RMSE	0.0039	0.0070	0.0045	0.0035	0.0037	0.0033	**0.0032**
	CC	0.9916	0.9805	0.9891	0.9931	0.9924	0.9941	**0.9943**
	SAD	1.3260	4.6881	1.6308	1.4643	1.4835	1.3065	**1.2913**
	ERGAS	2.1387	4.3320	2.3877	1.9495	2.0077	1.7861	**1.7487**
	PSNR	48.1462	43.1590	46.9921	49.1981	48.5289	49.7171	**49.9097**
	SSIM	0.9882	0.9675	0.9864	0.9911	0.9901	0.9909	**0.9913**
nachal_0823-1110	RMSE	0.0118	0.0191	0.0125	0.0109	0.0105	0.0105	**0.0104**
	CC	0.9906	0.9807	0.9898	0.9923	**0.9930**	0.9926	0.9927
	SAD	0.8890	3.8973	1.1163	1.0485	1.1369	0.8754	**0.8588**
	ERGAS	1.7652	2.9279	1.8462	1.6043	1.5464	1.5553	**1.5424**
	PSNR	38.5273	34.3872	38.0958	39.2324	39.5444	39.5995	**39.6698**
	SSIM	0.9380	0.8336	0.9350	0.9464	**0.9504**	0.9469	0.9475
objects_0924-1622	RMSE	0.0075	0.0177	0.0082	0.0065	0.0060	0.0060	**0.0059**
	CC	0.9955	0.9844	0.9947	0.9966	0.9970	0.9972	**0.9972**
	SAD	0.6125	3.1185	0.7260	0.6889	0.6932	0.5960	**0.5932**
	ERGAS	0.8091	1.9650	0.8809	0.7088	0.6709	0.6454	**0.6385**
	PSNR	42.5413	35.0227	41.7421	43.7816	44.4429	44.4784	**44.5719**
	SSIM	0.9742	0.8733	0.9719	0.9781	**0.9802**	0.9784	0.9784
rsh_0406-1427	RMSE	0.0114	0.0174	0.0124	0.0108	0.0106	0.0101	**0.0101**
	CC	0.9844	0.9705	0.9132	0.9865	0.9871	0.9875	**0.9875**
	SAD	1.3498	3.9978	0.9824	1.4253	1.4490	1.2680	**1.2614**
	ERGAS	2.0627	3.6436	1.8803	1.9742	1.9588	**1.8367**	1.8380
	PSNR	38.8802	35.1741	38.1600	39.2967	39.5124	39.8786	**39.8965**
	SSIM	0.9184	0.8595	0.9132	0.9319	**0.9363**	0.9326	0.9329
sami_0331-1019	RMSE	0.0084	0.0214	0.0093	0.0070	0.0067	0.0066	**0.0065**
	CC	0.9979	0.9927	0.9974	0.9986	0.9986	0.9987	**0.9988**
	SAD	0.9583	4.7497	1.2969	1.0803	1.1481	0.9045	**0.8710**
	ERGAS	1.2657	4.3012	1.4115	1.0968	1.1382	0.9854	**0.9710**
	PSNR	41.5661	33.3832	40.5942	43.0532	43.5320	43.6348	**43.7964**
	SSIM	0.9623	0.8559	0.9581	0.9701	**0.9782**	0.9707	0.9728

Concerning the fusion-based HSI SR methods, their performance is severely worse than the other methods. This phenomenon indicates their high dependence on the HR RGB images. To better exhibit this high dependence, their performance with the auxiliary HR RGB images provided has also been listed in Table II. It is noticed that the performance of both methods has been greatly improved. Furthermore, the improvement on the spatial measurement PSNR is much greater than that on the spectral improvement. This observation validates that the HR RGB images play important roles in preserving the spatial information for the reconstruction process. Compared with the single-to-single-based methods, TrsfrLea and DSSE, both methods have incorporated the CNMF to make a further exploitation of the spectral information. However, the CNMF operation rebuilds the spectral curves via the dictionary and ignores the spectral correlation existed in the input HSI, making their worse performance on the spectral preservation.

In addition, considering that the images from the ICVL data set are with large size, these HSIs are downsampled by a ratio of 8 to further validate the performance of the proposal. The quantitative experimental data are displayed in Table III, and the performance of the fusion-based NSSR and LTTR with the HR RGB images given has also been exhibited at the right region of Table II. It is noted that for most of the test HSIs, the proposal still achieves a better spatial reconstruction performance than the other methods. Meanwhile, it is noted that the DSSE method always achieves the best SSIM measurement, which demonstrates the structure similarity between the reconstructed HSI and the reference HSI. However, according to the SAD measurement, the HSI

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TABLE II
PERFORMANCES OF THE FUSION-BASED METHODS ON THE ICVL DATA SET WHEN THE RGB IMAGES ARE PROVIDED

image	method	4x						8x					
		RMSE	CC	SAD	ERGAS	PSNR	SSIM	RMSE	CC	SAD	ERGAS	PSNR	SSIM
bgu_0403-1511	NSSR	0.0205	0.9906	3.1699	2.9135	33.7761	0.8993	0.0159	0.9852	3.3475	1.8154	35.9717	0.9683
	LTTR	0.0086	0.9960	1.1078	1.0323	41.2972	0.9834	0.0085	0.9945	1.2656	0.6421	41.4463	0.9832
gavyam_0823-0933	NSSR	0.0160	0.9895	3.2043	3.0567	35.9331	0.9047	0.0100	0.9961	2.8967	1.0977	39.9621	0.9841
	LTTR	0.0063	0.9973	0.7638	0.8852	44.0053	0.9914	0.0060	0.9966	0.8773	0.5214	44.4571	0.9918
hill_0325-1235	NSSR	0.0049	0.9883	4.1334	3.8055	46.1735	0.9812	0.0105	0.9913	2.8017	1.0283	39.6172	0.9836
	LTTR	0.0027	0.9942	1.3613	1.6495	51.5065	0.9961	0.0047	0.9979	0.8336	0.4130	46.5907	0.9930
lst_0408-0950	NSSR	0.0165	0.9850	3.7153	2.5495	35.6713	0.9003	0.0032	0.9888	2.8829	1.7354	49.8285	0.9924
	LTTR	0.0054	0.9980	0.7487	0.8185	45.3237	0.9910	0.0026	0.9929	1.5390	1.0210	51.5939	0.9961
nachal_0823-1110	NSSR	0.0154	0.9882	4.1561	2.5557	36.2701	0.8778	0.0089	0.9967	2.6463	0.8371	40.9928	0.9790
	LTTR	0.0054	0.9982	0.9510	0.8569	45.4178	0.9905	0.0050	0.9983	1.0993	0.4639	45.9592	0.9918
objects_0924-1622	NSSR	0.0158	0.9873	3.3356	1.8730	36.0011	0.8866	0.0083	0.9968	2.1812	0.5780	41.5947	0.9860
	LTTR	0.0042	0.9984	0.6358	0.4864	47.5943	0.9922	0.0042	0.9979	0.6960	0.2794	47.4465	0.9927
rsh_0406-1427	NSSR	0.0125	0.9815	4.2236	3.2125	38.0588	0.9158	0.0103	0.9861	3.7999	1.8077	39.7401	0.9745
	LTTR	0.0058	0.9947	1.4535	1.4128	44.7809	0.9868	0.0054	0.9940	1.6388	0.8150	45.4007	0.9891
sami_0331-1019	NSSR	0.0195	0.9938	5.0013	4.4187	34.2154	0.8792	0.0120	0.9971	3.9707	2.0136	38.4187	0.9701
	LTTR	0.0049	0.9989	1.0382	0.8673	46.2487	0.9911	0.0049	0.9981	1.2708	0.5621	46.1745	0.9916

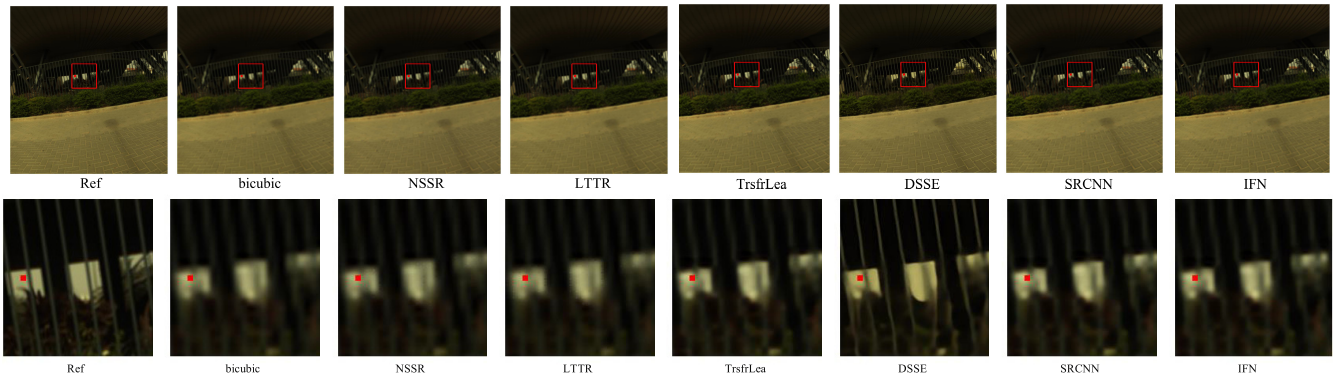


Fig. 5. Pseudogray image of the reconstructed HR "sami_0331-1019" HSIs with different methods.

reconstructed by the DSSE is always severally deviated from the reference HSI in the spectral dimension. In this way, the PSNR measurement that calculates the pixel deviation according to the MSE value is also being effected. When combing the experimental data of the NSSR and the LTTR methods in both Tables II and III, it is noticed that their performance still highly depends on the HR RGB images. Furthermore, when the scale is larger, the information these HR RGB images conveys is more critical for these fusion methods, and their performance are more severely influenced.

For qualitative analysis, Fig. 5 shows the RGB descriptions of the eight times reconstructed "sami_0331-1019" HSI via different methods. The first to the 11th bands whose spectra range from 400 to 500 nm are averaged and act as the B channel. The 11th to the 21st bands with the spectra ranging from 500 to 600 nm are averaged and act as the G channel, and the 21st to the 31st bands ranging from 600 to 700 nm are utilized to generate the R channel. Visually, it is noticed that both the SRCNN and the IFN can better recover the spatial details via their networks compared with the bicubic, NSSR, and LTTR methods. There are many inconsistent pixels in the HSIs reconstructed by the NSSR and the LTTR methods that are distinct from the background of the scene. Meanwhile, compared with the TrsfrLea and the SRCNN methods, there are some areas that the proposed IFN achieves a better

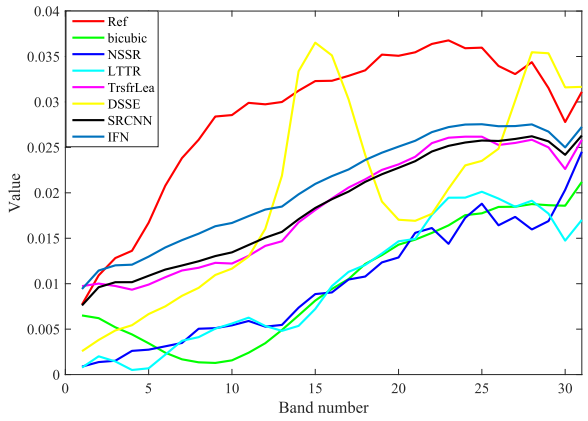


Fig. 6. Spectral curves of the randomly selected window pixel in the reconstructed HSIs.

boundary preserving performance. For example, in the red rectangle area, the top edge of the second window by the proposed IFN is sharper and more continuous than those by the SRCNN and the TrsfrLea methods. Furthermore, in the case of the HSI reconstructed by the DSSE method, its structure and edge are much more shaper than the other HSIs. However, this RGB image looks obviously different from the other

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TABLE III
PERFORMANCES OF DIFFERENT SR METHODS ON THE EIGHT TEST HSIS FROM THE ICVL DATA SET AT THE SCALING FACTOR OF 8

image	index	bicubic	NSSR [8]	LTTR [6]	TrsfrLea [17]	DSSE [19]	SRCNN [22]	IFN
bgu_0403-1511	RMSE	0.0313	0.0416	0.0322	0.0294	0.0306	0.0298	**0.0292**
	CC	0.9712	0.9545	0.9695	0.9747	0.9657	**0.9753**	0.9749
	SAD	1.4346	3.5251	1.7884	1.5005	2.5369	1.4092	**1.4057**
	ERGAS	1.5546	2.4274	1.5897	1.4507	1.7164	**1.4457**	1.4547
	PSNR	30.0944	27.6282	29.8347	30.6262	30.2805	30.5194	**30.6984**
	SSIM	0.7968	0.7757	0.7921	0.8133	**0.8281**	0.8135	0.8118
gavyam_0823-0933	RMSE	0.0247	0.0290	0.0252	0.0234	**0.0209**	0.0235	0.0229
	CC	0.9634	0.9607	0.9624	0.9680	0.9738	**0.9688**	0.9686
	SAD	0.8793	3.0587	1.1257	1.1214	1.4299	0.8856	**0.8785**
	ERGAS	1.5877	1.8500	1.6095	1.4846	**1.3524**	1.4700	1.4748
	PSNR	32.1360	30.7571	31.9565	32.6114	**33.5815**	32.5824	32.7917
	SSIM	0.8908	0.8808	0.8887	0.8968	**0.9159**	0.8967	0.8959
hill_0325-1235	RMSE	0.0185	0.0237	0.0188	0.0184	0.0225	0.0187	**0.0179**
	CC	0.9794	0.9725	0.9790	0.9802	0.9660	0.9806	**0.9806**
	SAD	0.9128	3.0042	1.0672	1.0795	3.0020	0.9128	**0.9096**
	ERGAS	1.3228	1.6564	1.3345	1.2901	1.6731	1.2930	**1.2864**
	PSNR	34.6748	32.5110	34.5042	34.7007	32.9544	34.5844	**34.9268**
	SSIM	0.8769	0.8680	0.8759	0.8821	0.8821	0.8822	**0.8822**
lst_0408-0950	RMSE	0.0079	0.0097	0.0082	0.0074	0.0076	0.0076	**0.0073**
	CC	0.9660	0.9583	0.9640	0.9715	0.9715	**0.9718**	0.9715
	SAD	1.6756	3.2182	1.9523	1.7710	2.0946	1.7693	**1.6841**
	ERGAS	2.1153	2.6381	2.1705	1.9449	2.0275	2.0115	**1.9240**
	PSNR	42.0742	40.3033	41.7175	42.6719	42.4403	42.3744	**42.7848**
	SSIM	0.9669	0.9615	0.9658	0.9711	**0.9726**	0.9705	0.9711
nachal_0823-1110	RMSE	0.0194	0.0223	0.0197	0.0187	0.0186	0.0191	**0.0185**
	CC	0.9752	0.9724	0.9746	0.9773	**0.9789**	0.9776	0.9775
	SAD	1.2481	3.0114	1.4070	1.3764	1.7954	1.2421	**1.2392**
	ERGAS	1.4532	1.6468	1.4716	1.3851	**1.3713**	1.3881	1.3829
	PSNR	34.2322	33.0527	34.0972	34.5476	34.6043	34.3829	**34.6651**
	SSIM	0.8875	0.8699	0.8860	0.8925	**0.8988**	0.8927	0.8924
objects_0924-1622	RMSE	0.0144	0.0174	0.0147	0.0137	**0.0123**	0.0143	0.0134
	CC	0.9834	0.9813	0.9827	0.9853	**0.9882**	0.9857	0.9856
	SAD	0.7750	2.2739	0.8620	0.8259	1.0010	0.7637	**0.7612**
	ERGAS	0.7770	0.9504	0.7928	0.7312	**0.6576**	0.7567	0.7284
	PSNR	36.8604	35.1942	36.6353	37.2929	**38.2341**	36.8728	37.4531
	SSIM	0.9453	0.9372	0.9441	0.9479	**0.9520**	0.9478	0.9476
rsh_0406-1427	RMSE	0.0182	0.0237	0.0187	0.0179	0.0199	0.0178	**0.0175**
	CC	0.9609	0.9491	0.9594	0.9630	0.9611	**0.9641**	0.9639
	SAD	1.9301	4.3395	2.3184	1.9934	2.8906	**1.8792**	1.8806
	ERGAS	1.6390	2.3757	1.6741	1.6068	2.0762	**1.5666**	1.5754
	PSNR	34.7965	32.5120	34.5712	34.9270	34.0085	34.9797	**35.1424**
	SSIM	0.8455	0.8286	0.8431	0.8528	**0.8578**	0.8532	0.8529
sami_0331-1019	RMSE	0.0172	0.0242	0.0176	0.0159	0.0161	0.0164	**0.0158**
	CC	0.9912	0.9887	0.9908	0.9925	0.9922	**0.9927**	0.9926
	SAD	1.4494	4.4012	1.7017	1.4834	2.9888	1.3868	**1.3850**
	ERGAS	1.3016	2.4048	1.3278	1.1980	1.2822	1.2062	**1.1960**
	PSNR	35.2678	32.3190	35.0794	35.9521	35.8641	35.7198	**36.0101**
	SSIM	0.8991	0.8762	0.8971	0.9083	**0.9237**	0.9082	0.9087

images in the hue perspective. It is induced that the spectral information of the HSI is not well preserved via the DSSE method. To further validate this induction, we have randomly selected one pixel in the window from all the reconstructed HSIs and plotted their spectral curves, which have been demonstrated in Fig. 6. According to the curves shown in Fig. 6, it is noticed that the spectral curve of the DSSE method is severally deviated from the reference one. Figs. 5 and 6 have proved the effectiveness of the DSSE in recovering the structure, but neglecting the spectral preservation, which is consistent with the data in Tables I and III. In this way, for the "gavyam_0823-0933" HSI, the main body of the scene is a bridge whose structure is composed of many boundaries, and the DSSE achieves better spatial improvement than the IFN because of its superiority in recovering edges. However, the SAD of the DSSE is still much larger than the proposed IFN.

The second data set is the Pavia Center. To further validate the generalization and effectiveness of the intrafusion strategy, the residual network RIFN with intrafusion operation is applied to this data set. Fig. 7 shows the visual exhibition of the four times reconstructed HSIs. Compared with the rectified area in the HSIs super-resolved via different methods, it is seen that both the IFN and the RIFN obtain an area with less mosaic effect than the SRCNN and the VDSR, respectively. Boundaries of the buildings reconstructed by the network with intrafusion are much clearer than that by the normal network. Meanwhile, the HSIs reconstructed by the bicubic, NSSR, and the LTTR methods are filled with mosaic or blur effects, which further validates the sensitivity of the fusion-based

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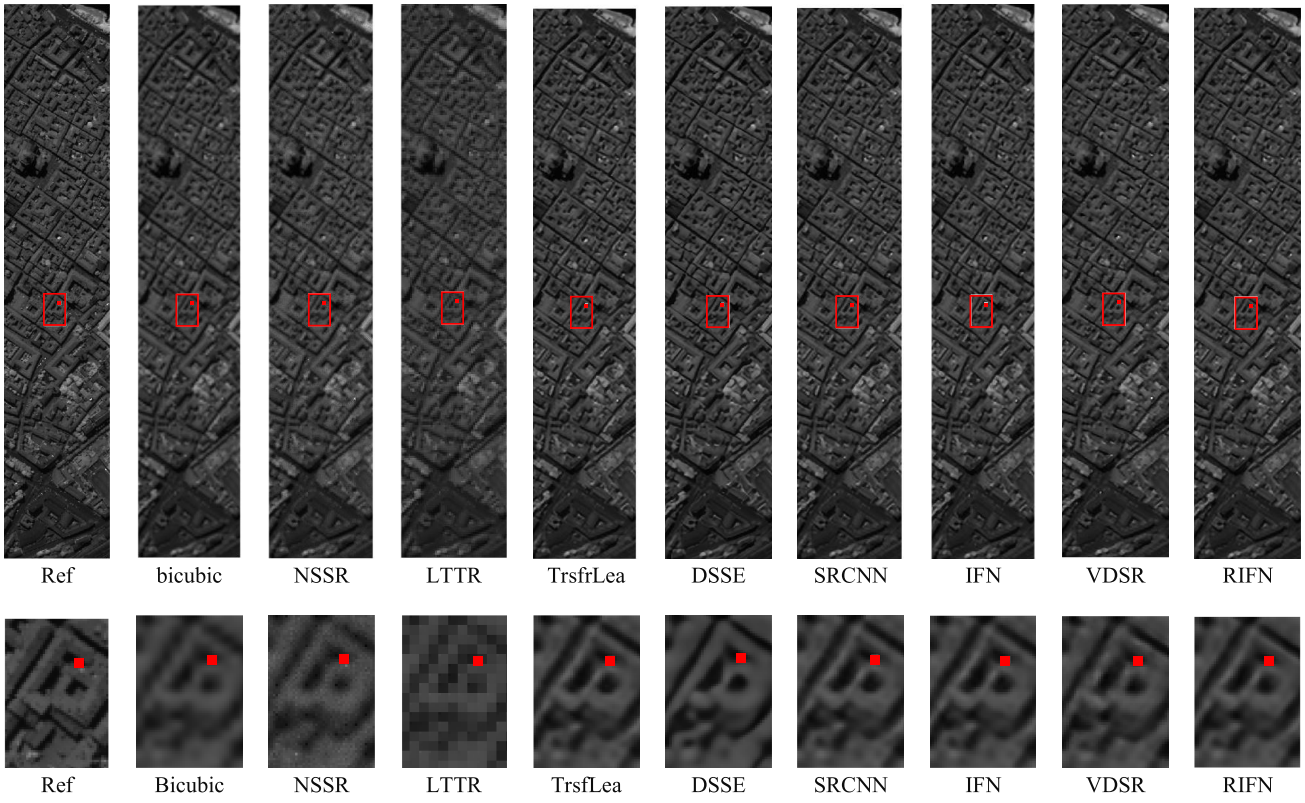


Fig. 7. Pseudogray image of the reconstructed HR Pavia Centers with different methods.

TABLE IV
PERFORMANCES OF THE HSI SR METHODS ON THE PAVIA CENTER

index	bicubic	NSSR [8]	LTTR [6]	TrsfrLea [17]	DSSE [19]	SRCNN [22]	IFN	VDSR [24]	RIFN
RMSE	0.0409	0.0447	0.0482	0.0368	0.0385	0.0368	**0.0367**	0.0371	0.0368
CC	0.8651	0.8530	0.8089	0.8892	0.8810	0.8896	**0.8897**	0.8878	0.8895
SAD	5.9761	8.3476	7.4293	5.7973	7.9335	5.7846	**5.7541**	6.1303	6.0213
ERGAS	6.8461	7.3121	8.0784	**6.1225**	7.5216	6.1510	6.1648	6.1628	6.1693
PSNR	27.7553	26.9902	26.3404	28.6776	28.3003	28.6890	**28.6950**	28.6103	28.6895
SSIM	0.7026	0.6998	0.6165	0.7602	**0.7625**	0.7597	0.7603	0.7571	0.7588

methods to the auxiliary HR RGB images. Concerning the HSI reconstrued by the DSSE method, some boundaries of the building are missing in the amplified red rectangle region, which produces some mistake information. Spectral curves of the red point in HSIs reconstructed via different methods are depicted in Fig. 8. It can be seen that the spectral curves of the IFN and RIFN are always closer to the reference one than those of the SRCNN and the VDSR. Among all the spectral curves, most bottoms and peaks of the curves reconstructed by the IFN and the RIFN are consistent with those of the reference one, which is an important prerequisite for their application in classification and detection. This indicates that the spectral information of HSIs is well preserved by our intrafusion strategy, and this intrafusion strategy can be further applied to various existed normal networks.

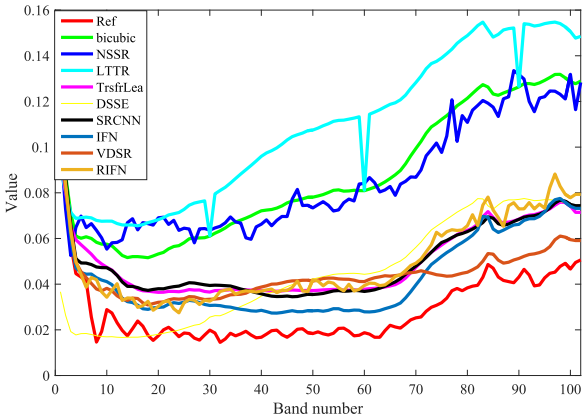


Fig. 8. Spectral curves of the selected pixel in the reconstructed Pavia Centers.

Quantitative experimental data of this HSI are shown in Table IV. Both the normal network SRCNN and the residual network VDSR achieve spatial and spectral improvement via the intrafusion strategy incorporated in the IFN and the RIFN. Meanwhile, the performance of the IFN and the RIFN outperforms the other methods simultaneously. Furthermore, spectral distortion of the DSSE method remains comparatively large, which is identical to the visual illustration in Fig. 8.

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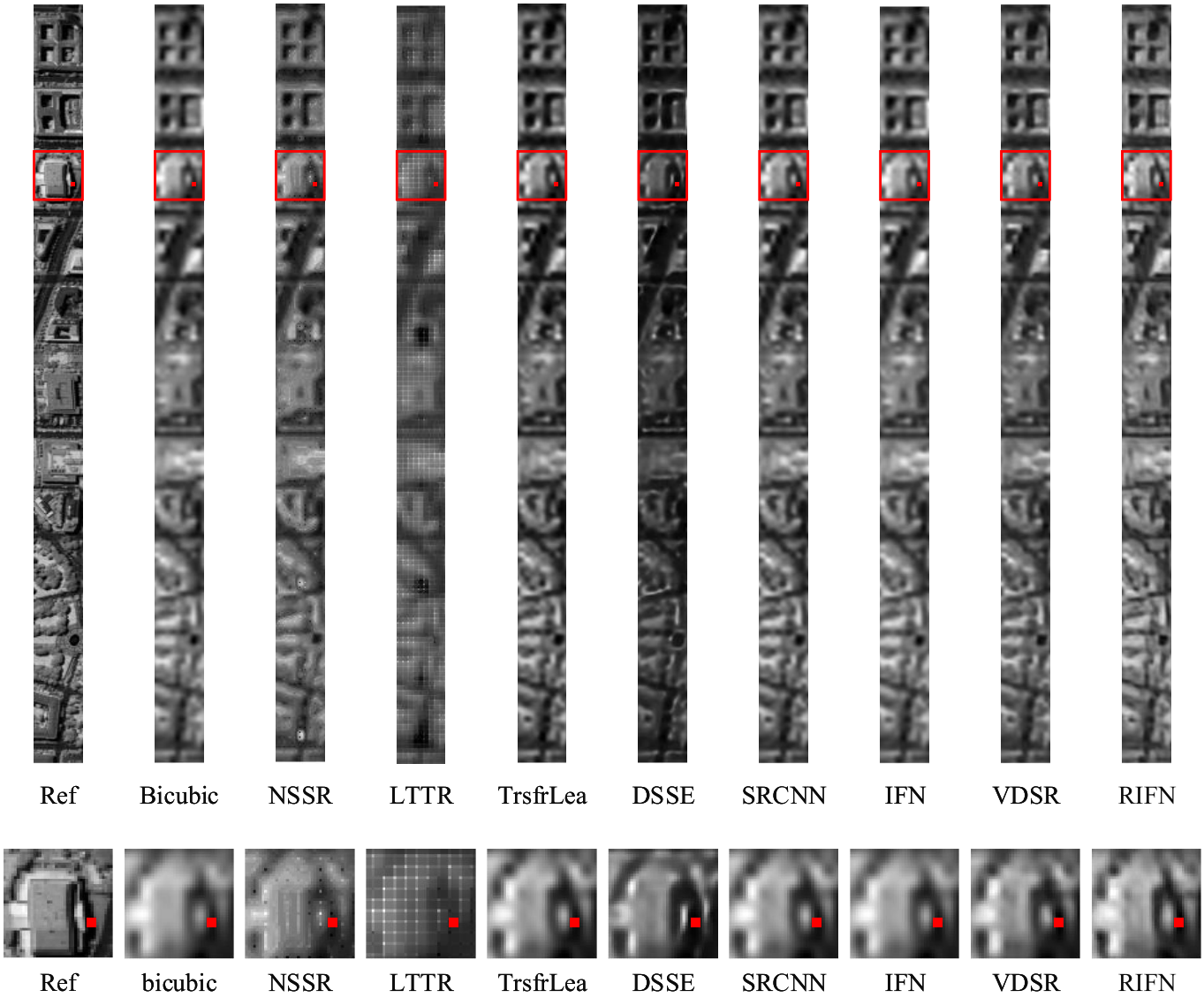


Fig. 9. Pseudogray image of the reconstructed HR Washington DC malls of different methods.

TABLE V
PERFORMANCES OF THE FUSION-BASED METHODS ON THE PAVIA CENTER AND THE WASHINGTON DC MALL WHEN THE RGB IMAGES ARE PROVIDED

Measurement	Pavia Center		Washington DC Mall	
	NSSR [8]	LTTR [6]	NSSR [8]	LTTR [6]
RMSE	0.0231	0.0444	0.0231	0.0178
CC	0.9477	0.8256	0.4660	0.6084
SAD	7.1598	7.6370	15.5332	11.9882
ERGAS	4.8230	7.7631	76.5361	55.5294
PSNR	32.7168	27.0483	32.7443	35.0036
SSIM	0.9126	0.6443	0.8987	0.8855

The performance of the NSSR and the LTTR with the auxiliary HR RGB images is also evaluated and presented in the left region of Table V. It is noticed that these two methods obtain great spatial improvement with the auxiliary HR RGB image, but the SAD is still comparatively high.

Fig. 9 shows the visual exhibition of the reconstructed Washington DC Malls via different methods. As described in the aforementioned section, this HSI is separated from the vertical direction to balance the sample diversity in the training, validation, and the testing set. Compared with the rectified area in the reconstructed HSIs, it is seen that the roofs of the building are better characterized via the SRCNN, IFN, VDSR, and RIFN than the other methods. The boundary reconstructed by the DSSE is sharper but is further deviated from the reference one. Among the SRCNN, IFN, VDSR, and the RIFN, the spatial improvements brought by the IFN and the RIFN are less when compared with the ICVL data set, and this may be caused by the limited training samples for this data set. Spectral curves of the red point in Washington DC Mall HSIs reconstructed via different methods are depicted in Fig. 10. It can be seen that the curve reconstructed by the IFN is the closest to the reference one.

Table VI shows the quantitative experimental data of the HSI. Both the IFN and the RIFN achieve slight spatial improvement over the SRCNN and the VDSR. It is noted that the SRCNN outperforms the VDSR, and the intrafusion strategy applied to both the SRCNN and the VDSR achieves

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TABLE VI
PERFORMANCES OF THE HSI SR METHODS ON THE WAHINGTON DC MALL

index	bicubic	NSSR [8]	LTTR [6]	TrsfrLea [17]	DSSE [19]	SRCNN [22]	IFN	VDSR [24]	RIFN
RMSE	0.0099	0.0231	0.0194	0.0097	0.0099	0.0096	**0.0096**	0.0097	0.0097
CC	0.8523	0.4660	0.5712	0.8525	0.8353	**0.8608**	0.8497	0.8561	0.8433
SAD	6.6933	15.5332	16.8661	6.5330	6.7966	6.4731	**6.4433**	6.6091	6.7083
ERGAS	50.3054	76.5361	682.8221	**40.6858**	42.7995	45.4692	50.1961	1731.3143	62.6374
PSNR	40.0712	32.7443	34.2567	40.2199	40.0556	40.3678	**40.3861**	40.2802	40.2803
SSIM	0.9398	0.8987	0.8810	0.9437	0.9415	0.9445	**0.9447**	0.9434	0.9431

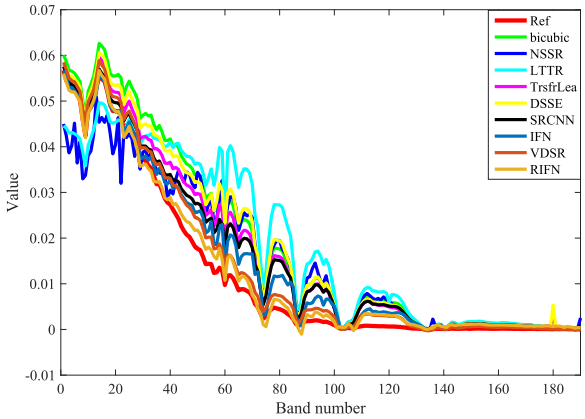


Fig. 10. Spectral curves of the selected pixel in the reconstructed Washington DC Malls.

the spatial improvement, making the IFN achieve the best spatial information among all the methods. In this way, it is foreseeable that this strategy can be applied to some other networks and also achieves spatial enhancement. The right region of Table V has also exhibited the performance of the fusion-based methods with the HR RGB image. Spatial improvement is still achieved, but the spectral distortion remains much higher than the single-to-single methods. The application of both the IFN and RIFN on Pavia Center and Washington DC Mall has demonstrated the generalization ability and effectiveness of our intrafusion strategy. In addition, this strategy is flexible and can be applied to more neural networks to achieve better performance.

IV. CONCLUSION

This article has presented a novel deep learning approach for HSI SR. Contrary to developing a wholly new architecture for HSI SR, this article proposes a strategy with three modules that can be applied to most neural networks and to compose an IFN. Both the spatial and spectral information can be simultaneously utilized in this integrated IFN, which provides a better SR performance than the normal network. The proposal in this article has opened up a new way for HSI SR. Experimental results and data analysis have demonstrated the effectiveness of this proposal.

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